

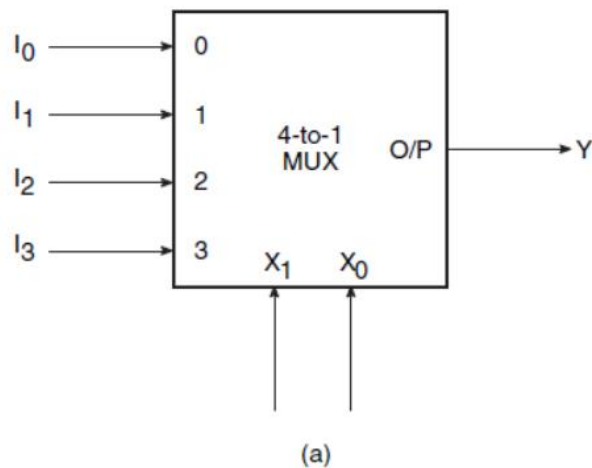
DIGITAL ELECTRONICS PREVIOUS YEAR QUESTIONS:

1. Design a full adder and implement it using suitable multiplexer.
2. Design the Boolean function $f(a,b,c) = \sum (3,7,4,5)$ and implement it using suitable multiplexer
3. Mention the difference between a DEMUX and a MUX.
4. What is a Multiplexer? Explain 4*1 Multiplexer. Also discuss the applications of multiplexers.
5. Implement the following function with multiplexer.
 - a. $F(A,B,C,D) = \sum (0,1,3,4,8,9,15)$
 - b. $F(A,B,C) = \sum (2,4,7)$
 - c. $F(x,y,z) = \sum (1,2,4,7)$
6. Implement a full adder circuit with multiplexer.

Hint: $S(x,y,z) = \sum (1,2,4,7)$ and $C(x,y,z) = \sum (3,5,6,7)$

Multiplexer

A *multiplexer* or *MUX*, also called a *data selector*, is a combinational circuit with more than one input line, one output line and more than one selection line. A multiplexer selects binary information present on any one of the input lines, depending upon the logic status of the selection inputs, and routes it to the output line. If there are n selection lines, then the number of maximum possible input lines is 2^n and the multiplexer is referred to as a 2^n -to-1 multiplexer or $2^n \times 1$ multiplexer. Figures 8.1(a) and (b) respectively show the circuit representation and truth table of a basic 4-to-1 multiplexer.



X_1	X_0	Y
0	0	I_0
0	1	I_1
1	0	I_2
1	1	I_3

(b)

Figure 8.1 (a) 4-to-1 multiplexer circuit representation and (b) 4-to-1 multiplexer truth table.

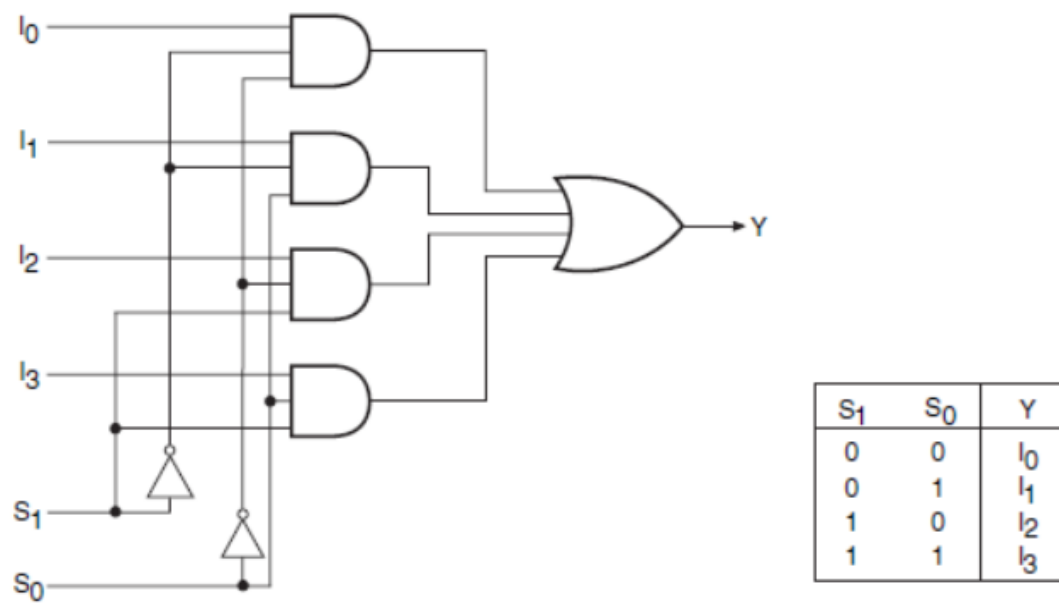


Figure 8.5 Logic diagram of a 4-to-1 multiplexer.

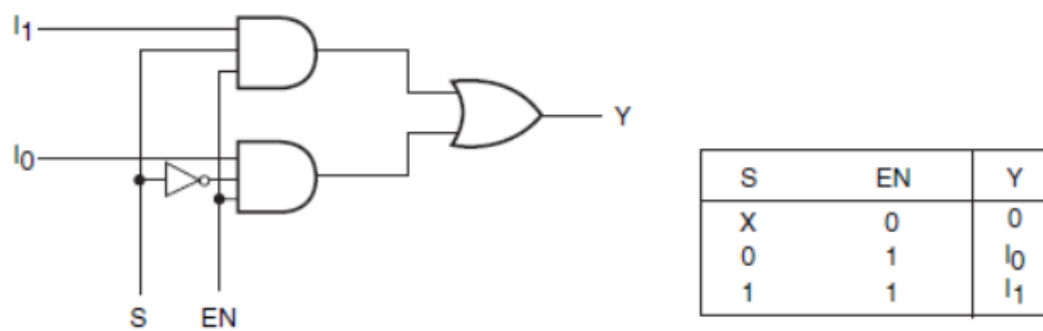


Figure 8.6 2-to-1 multiplexer with an ENABLE input.

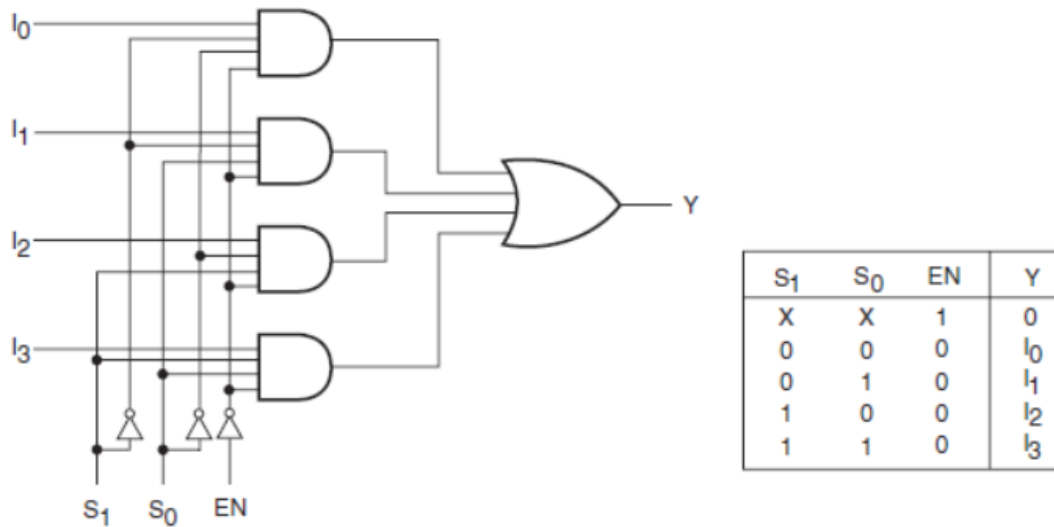


Figure 8.7 4-to-1 multiplexer with an ENABLE input.

Applications of Multiplexers

The multiplexer is one of the widely combinational logic circuits in digital systems. Some important applications of the multiplexers are listed below.

- Multiplexer is used as data selector device.
- Multiplexers are used in communication systems to increase the efficiency of the system.
- Multiplexers are used in telephone networks for integration of several audio signals on a single transmission line.
- To maintain large amount of data, multiplexers are also used in computer memory systems.
- Multiplexers are also employed in TV broadcasting systems.
- Multiplexers are used in satellite communication and GPS (Global Positioning System).
- Multiplexers are also used in PLC (Programmable Logic Control) systems, etc.

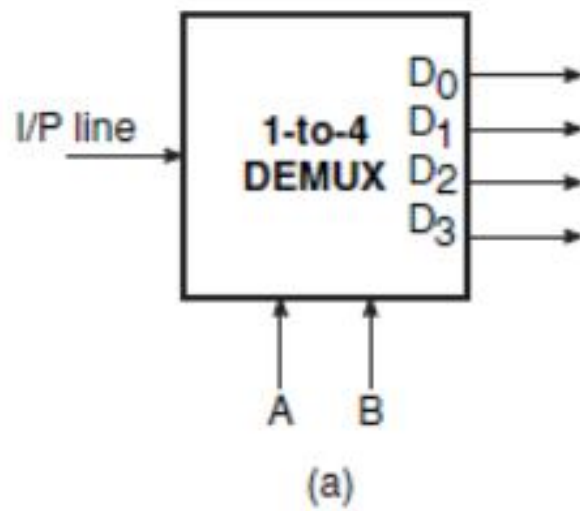
some other Applications are:

- **Communication System**
 - A communication system has both a communication network and a transmission system. By using a multiplexer, the efficiency of the communication system can be increased by allowing the transmission of data, such as audio and video data from different channels through single lines or cables.
- **Computer Memory**
 - Multiplexers are used in computer memory to maintain a huge amount of memory in the computers, and also to reduce the number of copper lines required to connect the memory to other parts of the computer.
- **Telephone Network**
 - In telephone networks, multiple audio signals are integrated on a single line of transmission with the help of a multiplexer.
- **Transmission from the Computer System of a Satellite**
 - The multiplexer is used to transmit the data signals from the computer system of a spacecraft or a satellite to the ground system by using a GSM satellite.

Demultiplexers

De-multiplexer is also a device with one input and multiple output lines. It is used to send a signal to one of the many devices. The main difference between a multiplexer and a de-multiplexer is that a multiplexer takes two or more signals and encodes them on a wire, whereas a de-multiplexer does reverse to what the multiplexer does.

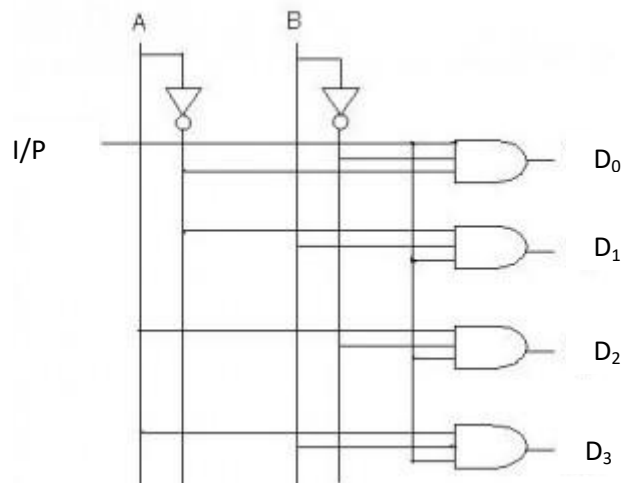
A demultiplexer is a combinational logic circuit with an input line, 2^n output lines and n select lines. It routes the information present on the input line to any of the output lines. The output line that gets the information present on the input line is decided by the bit status of the selection lines. Figure 8.18(a) shows the circuit representation of a 1-to-4 demultiplexer. Figure 8.18(b) shows the truth table of the demultiplexer when the input line is held HIGH.



I/P	Select		O/P			
	A	B	D ₀	D ₁	D ₂	D ₃
1	0	0	1	0	0	0
1	0	1	0	1	0	0
1	1	0	0	0	1	0
1	1	1	0	0	0	1

(b)

Figure 8.18 1-to-4 demultiplexer.



The main difference between multiplexer and demultiplexer is discussed below.

Multiplexer

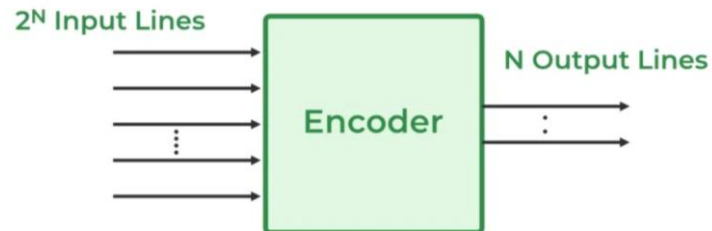
- A multiplexer (Mux) is a combinational circuit that uses several data inputs to generate a single output.
- Multiplexer includes several inputs and the single output
- A multiplexer is a data selector
- It is a digital switch
- It works on the principle of many to one
- The parallel to serial conversion is used in the multiplexer
- The multiplexer used in TDM (Time Division Multiplexing) is at the end of the transmitter
- In Multiplexer, control signals are used to choose the specific input that has to be sent at the output.
- The multiplexer is used to improve the efficiency of the communication system using transmission data like transmission of audio as well as video.
- The different types of multiplexers are 8-1 MUX, 16-1 MUX, and 32-1 MUX.
- In multiplexer, the set of selection lines are used to control the specific input

Demultiplexer

- A demultiplexer (Demux) is also a combinational circuit that uses single input that can be directed throughout several outputs.
- Demultiplexer includes single input and several outputs
- The demultiplexer is a data distributor
- It is a digital circuit
- It works on the principle of one-to-many
- The serial to parallel conversion is used in Demultiplexer
- The demultiplexer used in TDM (Time Division Multiplexing) is at the end of the receiver
- Demultiplexer uses the control signal to permit us to include several outputs.
- Demultiplexer gets the o/p signals from the Mux & changed them to the unique form at the end of the receiver.
- The different types of demultiplexers are 1-8 Demux, 1-16 Demux, 1-32 Demux.
- In demultiplexer, the selection of output line can be controlled through n-selection lines bit values.
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Encoders

An encoder is a multiplexer without its single output line. It is a combinational logic function that has 2^n (or fewer) input lines and n output lines, which correspond to n selection lines in a multiplexer. The n output lines generate the binary code for the possible 2^n input lines.



Let us take the case of an octal-to-binary encoder. Such an encoder would have eight input lines, each representing an octal digit, and three output lines representing the three-bit binary equivalent.

The truth table of such an encoder is given in Table 8.8. In the truth table, D0 to D7 represent octal digits 0 to 7. A, B and C represent the binary digits.

The eight input lines would have $2^8 = 256$ possible combinations. However, in the case of an octal-to-binary encoder, only eight of these 256 combinations would have any meaning. The remaining combinations of input variables are 'don't care' input combinations. Also, only one of the input lines at a time is in logic '1' state. Figure 8.15 shows the hardware implementation of the octal-to-binary encoder described by the truth table in Table 8.8.

This circuit has the shortcoming that it produces an all 0s output sequence when all input lines are in logic '0' state. This can be overcome by having an additional line to indicate an all 0s input sequence.

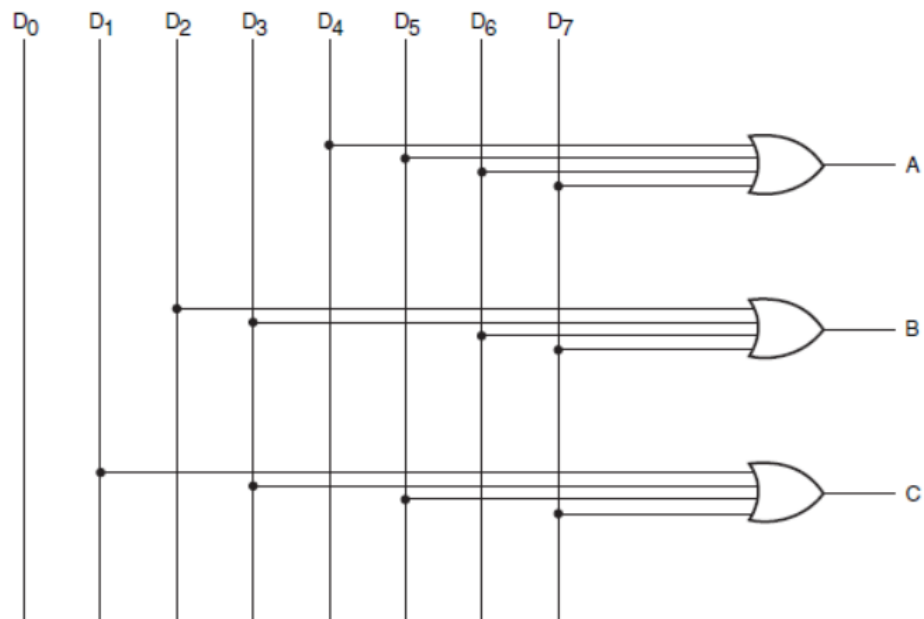


Figure 8.15 Octal-to-binary encoder.

Table 8.8 Truth table of an encoder.

D_0	D_1	D_2	D_3	D_4	D_5	D_6	D_7	A	B	C
1	0	0	0	0	0	0	0	0	0	0
0	1	0	0	0	0	0	0	0	0	1
0	0	1	0	0	0	0	0	0	1	0
0	0	0	1	0	0	0	0	0	1	1
0	0	0	0	1	0	0	0	1	0	0
0	0	0	0	0	1	0	0	1	0	1
0	0	0	0	0	0	1	0	1	1	0
0	0	0	0	0	0	0	1	1	1	1